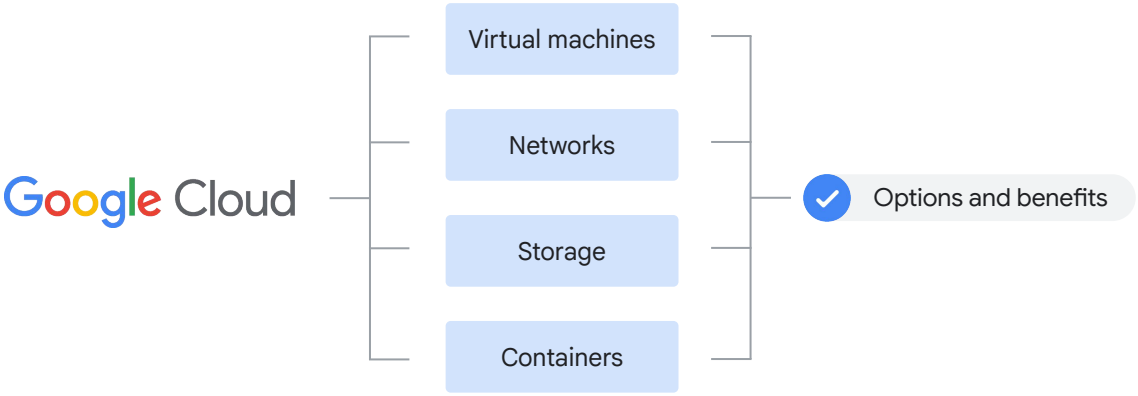
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Applications in the Cloud

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So far in this course ...



Google Cloud

So far in this course, we've provided an introduction to Google Cloud and explored the options and benefits related to using virtual machines, networks, storage, and containers in the Cloud.

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Applications in the Cloud

01 Cloud Run

02 Cloud Run functions



Google Cloud

In this final section of the course, we'll turn our attention to developing applications in the cloud by exploring Cloud Run and Cloud Run functions.

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Applications in the Cloud

01 Cloud Run

02 Cloud Run functions

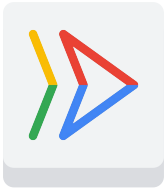


Google Cloud

Let’s begin with Cloud Run.

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Cloud Run is managed serverless computing



- A managed compute platform that can run stateless containers
- Serverless, removing the need for infrastructure management
- Built on Knative, an open API and runtime environment built on Kubernetes
- Can automatically scale up and down from zero almost instantaneously, charging only for the resources used

Google Cloud

Cloud Run is a managed compute platform that lets you run stateless containers via web requests or Pub/Sub events.

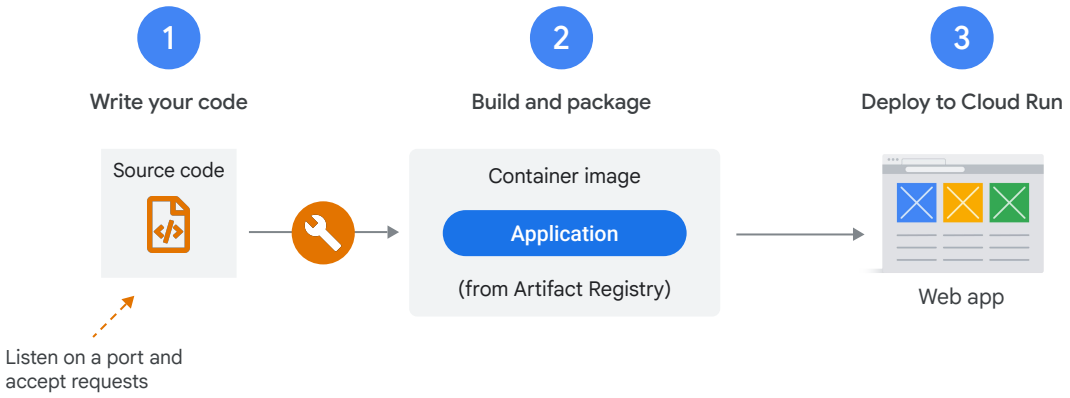
Cloud Run is serverless. That means it removes all infrastructure management tasks so you can focus on developing applications.

It is built on Knative, an open API and runtime environment built on Kubernetes. It can be fully managed on Google Cloud, on Google Kubernetes Engine, or anywhere Knative runs.

Cloud Run is fast. It can automatically scale up and down from zero almost instantaneously, and it charges only for the resources used, calculated down to the nearest 100 milliseconds, so you'll never pay for over-provisioned resources.

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Google Cloud Run workflow is a three-step process



Google Cloud

The Cloud Run developer workflow is a straightforward three-step process.

- First, you write your application using your favorite programming language. This application should start a server that listens for web requests.
- Second, you build and package your application into a container image.
- Third, the container image is pushed to Artifact Registry, where Cloud Run will deploy it.

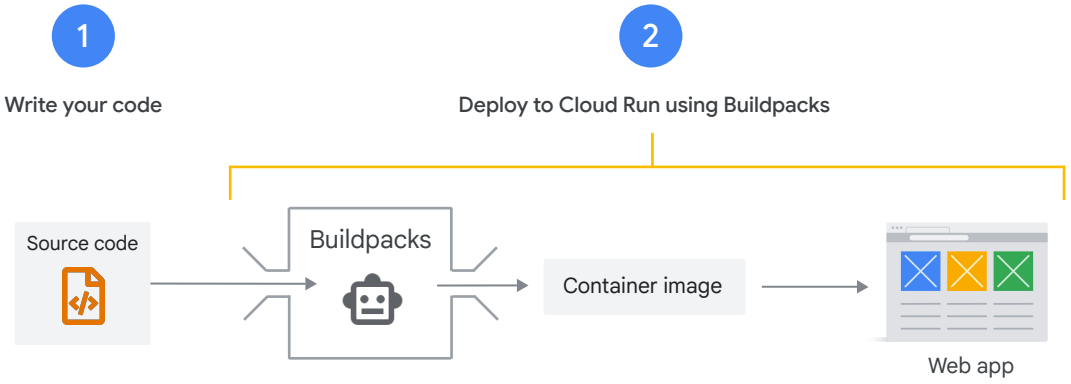
Once you've deployed your container image, you'll get a unique HTTPS URL back.

Cloud Run then starts your container on demand to handle requests, and ensures that *all incoming requests* are handled by dynamically adding and removing containers.

Cloud Run is serverless, which means that you, as a developer, can focus on building your application and not on building and maintaining the infrastructure that powers it.

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Cloud Run also has a source-based workflow



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For some use cases, a container-based workflow is great, because it gives you a great amount of transparency and flexibility. Sometimes, you're just looking for a way to turn source code into an HTTPS endpoint, and you want your vendor to make sure it your container image is secure, well-configured and built in a consistent way.

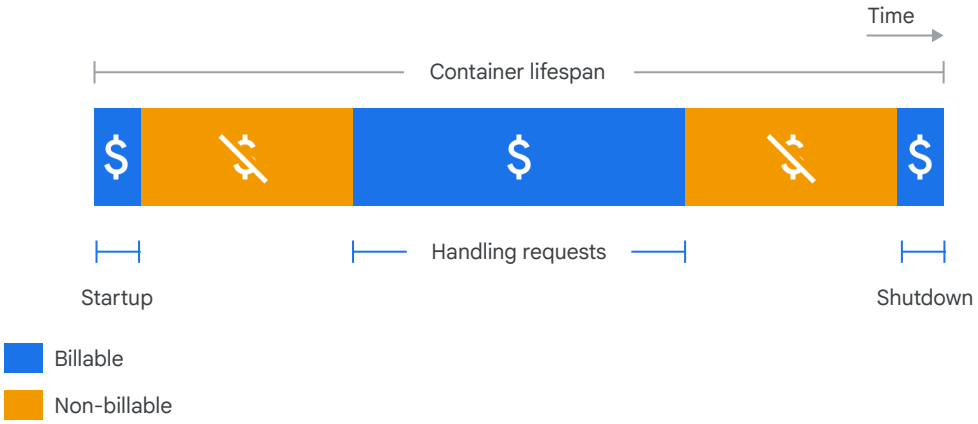
With Cloud Run, you can do both. You can use a container-based workflow, as well as a source-based workflow.

The source-based approach will deploy source code, instead of a container image. Cloud Run then builds the source and packages the application into a container image.

Cloud Run does this using Buildpacks - an open source project.

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You are only charged when your container handles requests



Google Cloud

The pricing model on Cloud Run is unique; as you only pay for the system resources you use while a container is handling web requests, with a granularity of 100ms, and when it is starting or shutting down.

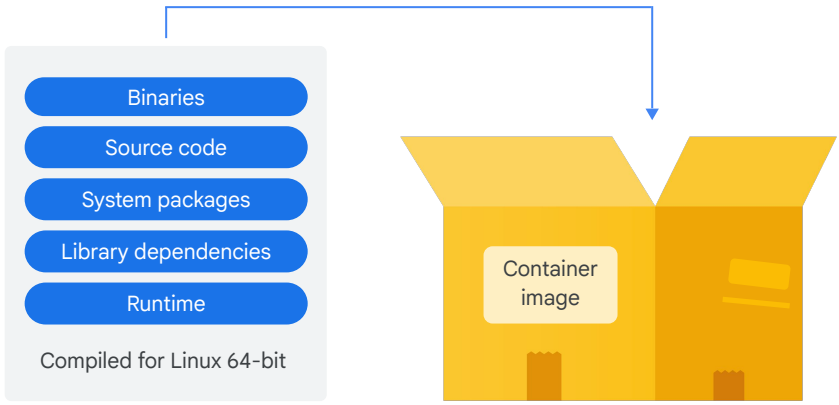
You do not pay for anything if your container does not handle requests. Additionally, there is a small fee for every one million requests you serve.

The price of container time increases with CPU and memory. A container with more vCPU and memory is more expensive. Today, Cloud run can allocate up to 4 vCPUs and 8GB of memory.

Most of the other compute products (such as Compute Engine), charge for servers as long as they are running, even if you are not using them. That means you're often paying for idle server capacity.

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Cloud Run can run any binary



Google Cloud

You can use Cloud Run to run any binary, as long as it is compiled for Linux sixty-four bit.

Now, this means you can use Cloud Run to run web applications written using popular languages, such as: Java, Python, Node.js, PHP, Go, and C++.

And you can also run code written in less popular languages: Cobol, Haskell, and Perl.

As long as your app handles web requests, you're good to go.

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Applications in the Cloud

01 Cloud Run

02 Cloud Run functions

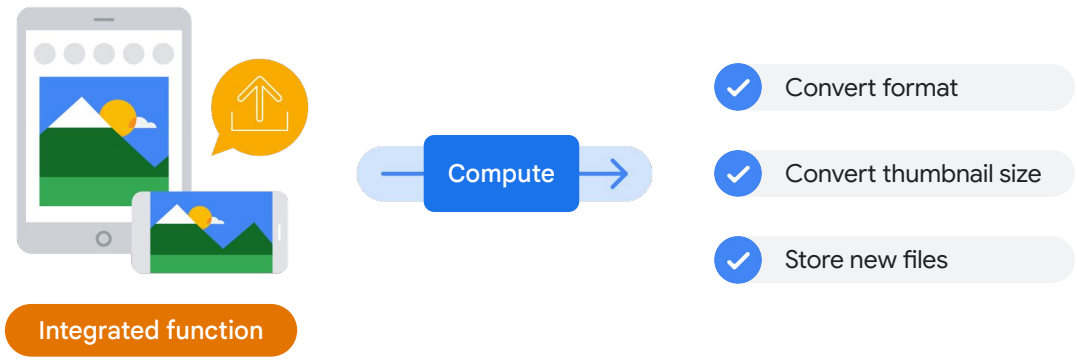


Google Cloud

And then there is **Cloud Run functions**.

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Integrated cloud functions handle application events



Google Cloud

Many applications contain event-driven parts. For example, an application that lets users upload images. When that event takes place, the image might need to be processed in a few different ways, like converting the image to a standard format, converting a thumbnail into different sizes, and storing each new file in a repository.

This function could be integrated into an application, but then you'd have to provide compute resources for it–whether it happens once a millisecond or once a day.

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Cloud Run functions allows your code to respond to events

- Lightweight, event-based, asynchronous compute solution
- Allows you to create small, single-purpose functions that respond to cloud events without the need to manage a server or a runtime environment
- Use these functions to construct applications from bite-sized business logic and connect and extend cloud services
- Billed to the nearest 100 milliseconds, and only while your code is running
- Supports writing source code in a number of programming languages, including Node.js, Python, Go, Java, .Net Core, Ruby, and PHP
- Events from Cloud Storage and Pub/Sub can trigger Cloud Run functions asynchronously, or use HTTP invocation for synchronous execution

Google Cloud

With **Cloud Run functions**, you write a single-purpose function that completes the necessary image manipulations, and then arrange for it to automatically run whenever a new image is uploaded.

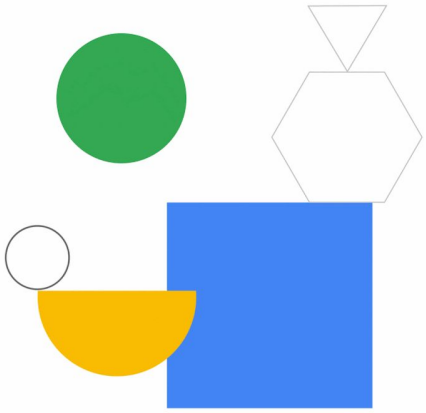
Cloud Run functions is a lightweight, event-based, asynchronous compute solution that allows you to create small, single-purpose functions that respond to cloud events without the need to manage a server or a runtime environment. These functions can be used to construct applications from bite-sized business logic. Cloud Run functions can also be used to connect and extend cloud services. You are billed to the nearest 100 milliseconds, but only while your code is running.

Cloud Run functions supports writing source code in a number of programming languages. These include Node.js, Python, Go, Java, .Net Core, Ruby, and PHP. For more information about the supported specific versions, refer to the runtimes documentation.

Events from Cloud Storage and Pub/Sub can trigger Cloud Run functions asynchronously, or you can use HTTP invocation for synchronous execution.

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Module Quiz



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Quiz | Question 1

Question

Which scenario is best suited for using Cloud Run instead of Cloud Run functions?

A. Sending an email notification whenever a new document is added to a specific folder in Cloud Storage.

B. Hosting a dynamic web application that allows users to upload and share photos.

C. Resizing images on demand when requested by a user through a web interface.

D. Generating thumbnails for images uploaded to a Cloud Storage bucket.

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Quiz | Question 1

Answer

Which scenario is best suited for using Cloud Run instead of Cloud Run functions?

A. Sending an email notification whenever a new document is added to a specific folder in Cloud Storage.

B. Hosting a dynamic web application that allows users to upload and share photos.



C. Resizing images on demand when requested by a user through a web interface.

D. Generating thumbnails for images uploaded to a Cloud Storage bucket.

Google Cloud

Which scenario is best suited for using Cloud Run instead of Cloud Run functions?

A. Sending an email notification whenever a new document is added to a specific folder in Cloud Storage.
Feedback: Incorrect. This event-driven task can be efficiently handled by Cloud Run functions, triggered by the upload event.

B. Hosting a dynamic web application that allows users to upload and share photos.
Feedback: Correct! This scenario requires a scalable web application with persistent connections and user interactions, making it ideal for Cloud Run's containerized environment and autoscaling capabilities.

C. Resizing images on demand when requested by a user through a web interface.
Feedback: Incorrect. While Cloud Run could handle this scenario, it’s an event-driven task that can be efficiently handled by Cloud Run functions, triggered by the upload event.

D. Generating thumbnails for images uploaded to a Cloud Storage bucket.
Feedback: Incorrect. This event-driven image processing task is well-suited for Cloud Run functions.

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Quiz | Question 2

Question

Which of these statements about Cloud Run functions are correct? Select three.

Cloud Run functions:

A. Is a scalable functions-as-a-service platform.

B. Can be used to extend Cloud services.

C. Require servers or VMs to be provisioned.

D. Is integrated with Cloud Logging.

E. Can only be invoked by sending HTTP requests.

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Quiz | Question 2

Answer

Which of these statements about Cloud Run functions are correct? Select three.

Cloud Run functions:

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E. Can only be invoked by sending HTTP requests.

Google Cloud

Cloud Run functions:

A: Is a scalable functions-as-a-service platform.
Correct! Cloud Run functions is a service that enables you to build single-purpose functions that can automatically scale based on the number of requests to your function.

B: Can be used to extend Cloud services.
Correct! Cloud Run functions can be used as a connective layer to connect and extend other cloud services and create a serverless application.

C: Require servers or VMs to be provisioned.
Incorrect. Cloud Run functions is a serverless environment, so you do not need to provision infrastructure or manage servers.

D: Is integrated with Cloud Logging.
Correct! Cloud Run functions is integrated with Google Cloud Observability logging and monitoring services to make it fully observable.

E: Can only be invoked by sending HTTP requests.
Incorrect. Functions are triggered by HTTP requests or events from your cloud infrastructure.

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Quiz | Question 3

Question

Why might a Google Cloud customer choose to use Cloud Run functions?

A. Their application contains event-driven code that they don't want to provision compute resources for.

B. Cloud Run functions is the primary way to run C++ applications in Google Cloud.

C. Cloud Run functions is a free service for hosting compute operations.

D. Their application has a legacy monolithic structure that they want to separate into microservices.

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Quiz | Question 3

Answer

Why might a Google Cloud customer choose to use Cloud Run functions?

A. Their application contains event-driven code that they don't want to provision compute resources for.



B. Cloud Run functions is the primary way to run C++ applications in Google Cloud.

C. Cloud Run functions is a free service for hosting compute operations.

D. Their application has a legacy monolithic structure that they want to separate into microservices.

Google Cloud

Why might a Google Cloud customer choose to use Cloud Run functions?

***A: Their application contains event-driven code that they don't want to provision compute resources for.**
Feedback: Correct!

B: Cloud Run functions is the primary way to run C++ applications in Google Cloud.
Feedback: Review the "Cloud Run functions" lecture.

C: Cloud Run functions is a free service for hosting compute operations.
Feedback: Review the "Cloud Run functions" lecture.

D: Their application has a legacy monolithic structure that they want to separate into microservices.
Feedback: Review the "Cloud Run functions" lecture.

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Lab Intro

Hello Cloud Run

The goal of this lab is for you to build a simple containerized application image and deploy it to Cloud Run.

